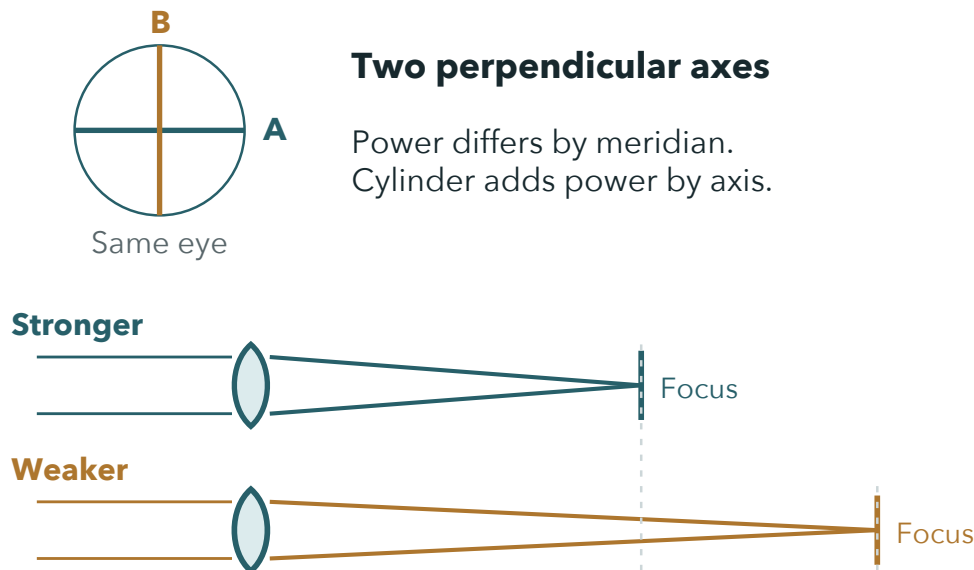


Optical meridians and pinhole

Different axes need different powers. Smaller apertures have a limit.

One eye, two principal powers

A | Why a cylinder changes the result



Two principal meridians focus at different distances. The line foci and orientation are schematic.

Read the refraction

Pattern	Focus problem	Correction
Spherical error	Similar power error in all meridians	Sphere shifts all meridians equally
Regular astigmatism	Different powers in perpendicular principal meridians	Cylinder or toric component adds axis-dependent power
Sphere plus cylinder	Overall defocus plus unequal meridional power	Correct both components

Fast check

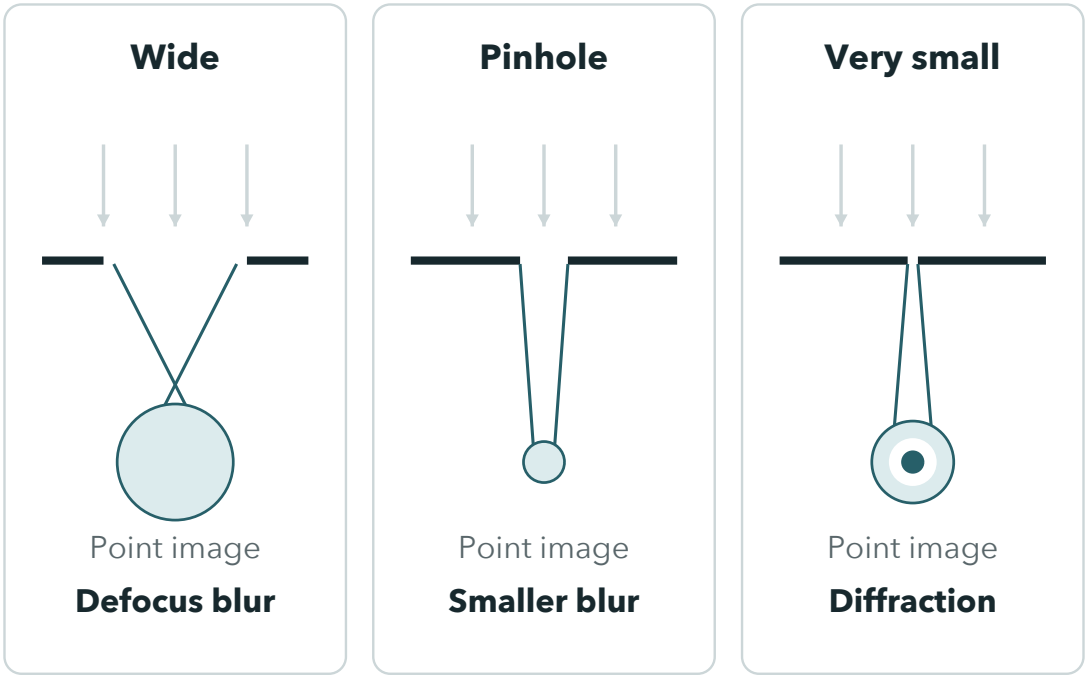
A spherical lens may improve one direction while leaving another blurred. Rotating a cylinder changes the meridian receiving its added power.

Figure reference

A: NEI refractive errors; supplied TBL2 optics. Original schematic.

Pinhole: improvement and its limit

B | Aperture versus image spread



Equal brightness is assumed for comparison. Blur profiles are conceptual, not measured point-spread functions.

Interpret the response

Finding	Mechanism	Limit
Clearer through a pinhole	Fewer peripheral rays; smaller defocus blur	Supports an optical component
Larger depth of focus	Greater tolerance of small focus errors	Does not correct the underlying refraction
Worse through a very tiny hole	Diffraction broadens the point image	Narrower is not always sharper

Fast check

A poor pinhole response does not identify one retinal or neural diagnosis. Illumination, aperture size, media clarity and the patient’s fixation still matter.

Figure reference

B: Webvision visual acuity. Original schematic.